

U23AI016

BDA

Assignment 8

```
cat > legacy.json << 'EOF'
```

```
{"id": 1, "info": {"zip": "395007", "city": "Surat"}}
```

```
{"id": 2, "info": {"zip": "400001", "city": "Mumbai"}}
```

```
{"id": 3, "info": {"zip": "110001", "city": "Delhi"}}
```

```
EOF
```

```
cat > avro_data.txt << 'EOF'
```

```
4,560001,active
```

```
5,700001,inactive
```

```
6,380001,active
```

```
EOF
```

```
cat > logs.txt << 'EOF'
```

```
101,999,click
```

```
102,999,view
```

```
103,999,click
```

```
104,201,click
```

```
105,202,view
```

```
EOF
```

```
cat > users.txt << 'EOF'
```

```
999,Bot_User
```

```
201,Alice
```

```
202,Bob
```

```
EOF
```

```
cat legacy.json
```

```
cat avro_data.txt
```

```
cat logs.txt
```

```
cat users.txt
```

```

u23ai016@cloudshell:~$ cat > avro_data.txt << 'EOF'
4,560001,active
5,700001,inactive
6,380001,active
EOF
u23ai016@cloudshell:~$ cat > logs.txt << 'EOF'
101,999,click
102,999,view
103,999,click
104,201,click
105,202,view
EOF
u23ai016@cloudshell:~$ cat > users.txt << 'EOF'
999,Bot_User
201,Alice
202,Bob
EOF
u23ai016@cloudshell:~$ cat legacy.json
cat avro_data.txt
cat logs.txt
cat users.txt
{"id": 1, "info": {"zip": "395007", "city": "Surat"}}
{"id": 2, "info": {"zip": "400001", "city": "Mumbai"}}
{"id": 3, "info": {"zip": "110001", "city": "Delhi"}}
4,560001,active
5,700001,inactive
6,380001,active
101,999,click
102,999,view
103,999,click
104,201,click
105,202,view
999,Bot_User
201,Alice
202,Bob
u23ai016@cloudshell:~$ █

```

1. Schema Evolution & Serialization Efficiency

1. Load both datasets using their respective loaders (JsonLoader and AvroStorage).
2. In the JSON relation, extract zip from the map and cast it to an int.
3. Perform a UNION of the two datasets, but ensure that the final schema is strictly (id, zip_code, source_flag).

Query:

```
cat > q1.pig << 'EOF'
```

```
-- Load JSON dataset using JsonLoader with explicit schema
```

```

json_data = LOAD 'legacy.json'
    USING PigStorage()
    AS (json_line:chararray);

-- Since PigStorage can't parse JSON maps, we use JsonLoader correctly
-- Reload with explicit schema passed inline
json_data2 = LOAD 'legacy.json'
    USING JsonLoader('id:int, info:map[]');

-- Extract zip from map and cast to int
json_clean = FOREACH json_data2 GENERATE
    id,
    (int) info#'zip' AS zip_code:int,
    'json' AS source_flag:chararray;

-- Load Avro substitute
avro_data = LOAD 'avro_data.txt'
    USING PigStorage(',')
    AS (id:int, zip_code:int, metadata:chararray);

-- Prepare avro relation
avro_clean = FOREACH avro_data GENERATE
    id,
    zip_code,
    'avro' AS source_flag:chararray;

-- Union both
merged = UNION ONSCHEMA json_clean, avro_clean;

```

DUMP merged;

EOF

pig -x local q1.pig

```
u23ai016@cloudshell:~$ cat > q1.pig << 'EOF'
-- Load JSON dataset
json_data = LOAD 'legacy.json'
            USING JsonLoader()
            AS (id:int, info:map[]);

-- Load Avro-like dataset (using PigStorage as substitute)
avro_data = LOAD 'avro_data.txt'
            USING PigStorage(',')
            AS (id:int, zip_code:int, metadata:chararray);

-- Extract zip from map and cast to int
json_clean = FOREACH json_data GENERATE
              id,
              (int) info#'zip' AS zip_code:int,
              'json' AS source_flag:chararray;

-- Prepare avro relation with same schema
avro_clean = FOREACH avro_data GENERATE
              id,
              zip_code,
              'avro' AS source_flag:chararray;

-- Union both datasets
merged = UNION ONSCHEMA json_clean, avro_clean;

DUMP merged;
EOF
u23ai016@cloudshell:~$
```

```
Input split[0]:
  Length = 50
  ClassName: org.apache.hadoop.mapreduce.lib.input.FileSplit
  Locations:
```

```
-----
2026-04-03 10:03:24,177 [LocalJobRunner Map Task Executor #0] INFO org.apache.pig.builtin.PigStorage - Using PigTextInputFormat
2026-04-03 10:03:24,180 [LocalJobRunner Map Task Executor #0] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.PigRec
```

```

HadoopVersion  PigVersion  UserId  StartedAt  FinishedAt  Features
2.7.3  0.17.0  u23ai016  2026-04-03 10:03:22  2026-04-03 10:03:24  UNION

Success!

Job Stats (time in seconds):
JobId  Maps  Reduces  MaxMapTime  MinMapTime  AvgMapTime  MedianMapTime  MaxReduceTime  MinReduceTime  AvgReduceTime  MedianReduceTime  Alias  Feature Out
job_local2004060075_0001  2  0  n/a  n/a  n/a  n/a  0  0  0  0  avro_clean,avro_data,json_clean,json_data2,merged  MAP_ONLY
/tmp923023914/tmp149740039,

Input(s):
Successfully read 3 records from: "file:///home/u23ai016/legacy.json"
Successfully read 3 records from: "file:///home/u23ai016/avro_data.txt"

Output(s):
Successfully stored 6 records in: "file:///tmp/temp923023914/tmp149740039"

Counters:
Total records written : 6
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local2004060075_0001

2026-04-03 10:03:24,560 [main] INFO  org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-03 10:03:24,565 [main] INFO  org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-03 10:03:24,570 [main] INFO  org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-03 10:03:24,585 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-04-03 10:03:24,591 [main] WARN  org.apache.pig.data.SchemaTupleBackend - SchemaTupleBackend has already been initialized
2026-04-03 10:03:24,616 [main] INFO  org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input paths to process : 2
2026-04-03 10:03:24,618 [main] INFO  org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 2

(4,560001,avro)
(5,700001,avro)
(6,380001,avro)
(1,395007,json)
(2,400001,json)
(3,110001,json)
2026-04-03 10:03:24,704 [main] INFO  org.apache.pig.Main - Pig script completed in 4 seconds and 917 milliseconds (4917 ms)
u23ai016@cloudshell:~$

```

2. Custom Data Partitioning & Skew-Join Mitigation

1. Provide the syntax for JOIN Logs BY user_id, Users BY id USING 'skewed';
2. Write a manual alternative where you split the Logs into two relations: Skewed_Logs (where user_id is the known bot ID) and Regular_Logs. How would you join these two relations differently to ensure the job completes?

QUERY:

```
cat > q2.pig << 'EOF'
```

```
-- Load both tables
```

```
Logs = LOAD 'logs.txt'
```

```
USING PigStorage(',')
```

```
AS (log_id:int, user_id:int, action:chararray);
```

```
Users = LOAD 'users.txt'
```

```
USING PigStorage(',')
```

```
AS (id:int, username:chararray);
```

```
-- Part A: Built-in Skewed Join
```

```
joined_skewed = JOIN Logs BY user_id, Users BY id  
    USING 'skewed';
```

```
DUMP joined_skewed;
```

```
EOF
```

```
pig -x local q2.pig
```

```
cat > q2b.pig << 'EOF'
```

```
-- Load both tables
```

```
Logs = LOAD 'logs.txt'  
    USING PigStorage(',')  
    AS (log_id:int, user_id:int, action:chararray);
```

```
Users = LOAD 'users.txt'  
    USING PigStorage(',')  
    AS (id:int, username:chararray);
```

```
-- Split logs into skewed and regular
```

```
SPLIT Logs INTO  
    Skewed_Logs IF (user_id == 999),  
    Regular_Logs IF (user_id != 999);
```

```
-- Join regular logs normally
```

```
regular_joined = JOIN Regular_Logs BY user_id, Users BY id;
```

```
-- Join skewed logs using replicated (map-side) join
```

```
skewed_joined = JOIN Skewed_Logs BY user_id, Users BY id
```

USING 'replicated';

-- Combine both results

final_result = UNION ONSCHEMA regular_joined, skewed_joined;

DUMP final_result;

EOF

pig -x local q2b.pig

```
(104,201,click,201,Alice)
(105,202,view,202,Bob)
(103,999,click,999,Bot_User)
(102,999,view,999,Bot_User)
(101,999,click,999,Bot_User)
2026-04-03 10:07:16,397 [main] INFO
u23ai016@cloudshell:~$
```

```

HadoopVersion  PigVersion  UserId  StartedAt  FinishedAt  Features
2.7.3  0.17.0  u23ai016  2026-04-03 10:09:16  2026-04-03 10:09:20  REPLICATED_JOIN,HASH_JOIN,UNION

Success!

Job State (time in seconds):
JobId  Maps  Reduces  MaxMapTime  MinMapTime  AvgMapTime  MedianMapTime  MaxReduceTime  MinReduceTime  AvgReduceTime  MedianReduceTime  Alias  Feature  Outputs
job_local12003212684_0001  1  0  n/a  n/a  n/a  n/a  0  0  0  0  Users  MULTI_QUERY,MAP_ONLY
job_local1680475477_0002  1  0  n/a  n/a  n/a  n/a  0  0  0  0  Skewed_Logs,final_result,skewed_joined  REPLICATED_JOIN,MAP_ONLY  file:/tmp/temp170
3474201/tmp1933685022,
job_local1943540799_0003  2  1  n/a  n/a  n/a  n/a  n/a  n/a  Regular_Logs,final_result,regular_joined  HASH_JOIN

Input(s):
Successfully read 5 records from: "file:///home/u23ai016/logs.txt"
Successfully read 3 records from: "file:///home/u23ai016/users.txt"

Output(s):
Successfully stored 5 records in: "file:/tmp/temp1703474201/tmp1933685022"

Counters:
Total records written : 5
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local12003212684_0001  ->  job_local1943540799_0003,job_local1813130658_0004,
job_local1680475477_0002 ->  job_local1943540799_0003,
job_local1943540799_0003 ->  job_local1813130658_0004,
job_local1813130658_0004

2026-04-03 10:09:20,816 [main] INFO  org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-03 10:09:20,816 [main] INFO  org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-03 10:09:20,822 [main] INFO  org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-03 10:09:20,837 [main] INFO  org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-03 10:09:20,841 [main] INFO  org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-03 10:09:20,843 [main] INFO  org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-03 10:09:20,854 [main] INFO  org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-03 10:09:20,858 [main] INFO  org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
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2026-04-03 10:09:20,873 [main] INFO  org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-03 10:09:20,875 [main] INFO  org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-03 10:09:20,878 [main] INFO  org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with processName=JobTracker, sessionId= - already initialized
2026-04-03 10:09:20,886 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-04-03 10:09:20,892 [main] WARN  org.apache.pig.data.SchemaTupleBackend - SchemaTupleBackend has already been initialized
2026-04-03 10:09:20,913 [main] INFO  org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input paths to process : 2
2026-04-03 10:09:20,914 [main] INFO  org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 2
(104,201,click,201,Alice,,)
(105,202,view,202,Bob,,)
(,,999,Bot_User,101,999,click)
(,,999,Bot_User,102,999,view)
(,,999,Bot_User,103,999,click)
2026-04-03 10:09:20,986 [main] INFO  org.apache.pig.Main - Pig script completed in 6 seconds and 972 milliseconds (6972 ms)
u23ai016@cloudshell:~$
```

```

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(104,201,click,201,Alice,,)
(105,202,view,202,Bob,,)
(,,999,Bot_User,101,999,click)
(,,999,Bot_User,102,999,view)
(,,999,Bot_User,103,999,click)
2026-04-03 10:09:20,986 [main] INFO  org.apache.pig.Main - Pig script completed in 6 seconds and 972 milliseconds (6972 ms)
u23ai016@cloudshell:~$
```